

TAKFA AGROMET, Thailand

WMO: 484010

Lat: 15.3498N

Lon: 100.5303E

Elev: 94

StdP: 100.20

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	19.0	9.0	7.2	21.2	10.8	8.2	22.6	4.4	23.6	4.1	24.3	0.7	N/A	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.9	38.1	25.8	37.1	26.1	36.1	26.0	27.7	33.5	27.5	33.2	27.2	32.6	1.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	22.3	30.0	26.2	21.8	29.7	25.9	21.5	29.6	89.3	33.9	88.0	33.2	86.8	32.8	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
3.9	3.4	3.0	DB	14.4	38.9	2.3	1.5	12.8	40.0	11.5	40.9	10.2	41.7	8.5	42.8	
			WB	12.0	28.7	2.4	0.6	10.3	29.1	8.9	29.4	7.5	29.7	5.8	30.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	26.1	28.3	30.1	31.2	30.6	29.6	28.9	28.6	28.4	28.2	27.7	26.1
	DBStd	2.24	2.26	1.74	2.09	1.91	1.77	1.21	1.22	0.89	0.98	1.07	1.46	2.20
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6801	499	512	622	636	638	589	585	577	551	564	529	498
	CDD18.3	3759	241	278	363	386	380	339	327	319	301	306	279	240
	CDH23.3	43118	2324	3167	4710	5257	4937	4106	3691	3431	3166	3204	2886	2240
	CDH26.7	19658	936	1496	2498	3010	2593	1886	1540	1339	1163	1199	1143	854
Wind	WSAvg	0.7	0.8	0.7	0.8	0.8	0.7	0.7	0.6	0.5	0.4	0.6	1.0	1.2
Precipitation	PrecAvg	1161	8	10	43	74	150	132	138	176	255	144	26	6
	PrecMax	1506	35	53	136	164	258	198	245	289	381	414	104	46
	PrecMin	805	0	0	0	10	33	41	58	88	155	8	0	0
	PrecStd	168	9	12	33	44	56	45	44	44	60	75	28	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.5	36.1	38.1	39.9	39.1	37.1	36.6	34.5	34.1	34.3	34.2	34.4
		MCWB	23.4	24.8	24.3	25.5	26.9	26.5	26.1	26.7	26.7	26.0	25.5	23.9
	2%	DB	33.3	35.1	37.0	38.7	37.6	35.6	34.4	33.5	33.3	33.3	33.4	33.3
		MCWB	23.3	24.5	24.9	25.8	26.6	26.5	26.2	26.5	26.6	25.9	25.2	24.1
	5%	DB	32.4	34.1	36.0	37.4	36.4	34.4	33.4	32.7	32.5	32.6	32.7	32.3
		MCWB	22.8	24.2	25.1	26.2	26.6	26.5	26.2	26.3	26.5	25.7	24.8	23.7
	10%	DB	31.4	33.2	34.9	36.2	35.1	33.3	32.3	32.0	31.7	31.8	31.8	31.1
		MCWB	22.5	24.0	25.2	26.2	26.6	26.4	26.2	26.3	26.4	25.7	24.4	23.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.0	26.9	27.4	28.2	28.4	27.8	27.6	27.7	27.7	27.5	27.2	26.2
		MCDB	31.1	34.2	34.4	35.3	34.7	33.4	32.3	32.2	31.9	31.7	32.3	31.6
	2%	WB	25.2	26.2	26.8	27.6	27.8	27.3	27.1	27.2	27.3	27.1	26.7	25.4
		MCDB	30.2	32.6	33.5	34.7	33.9	32.8	32.0	31.7	31.5	31.2	31.4	30.8
	5%	WB	24.7	25.6	26.4	27.2	27.4	27.1	26.8	26.9	27.0	26.7	26.1	24.7
		MCDB	29.6	31.6	32.9	34.1	33.4	32.4	31.5	31.3	31.2	30.6	30.3	30.0
	10%	WB	24.0	25.2	26.0	26.9	27.0	26.7	26.5	26.6	26.7	26.4	25.5	24.0
		MCDB	28.7	30.6	32.4	33.5	32.9	31.7	31.0	30.7	30.5	30.0	29.5	29.3
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.3	9.9	9.9	8.6	7.8	7.3	7.3	6.9	7.3	8.3	9.4
		MCDBR	11.7	11.3	11.3	11.2	10.2	9.2	8.6	8.2	8.0	8.4	9.4	10.4
	5% WB	MCWBR	4.4	3.8	3.1	2.9	2.6	2.5	2.5	2.7	2.8	2.7	3.3	3.8
		MCWBR	9.1	9.9	10.1	10.1	9.2	8.1	7.6	7.7	7.5	7.5	8.4	9.4
Clear-Sky Solar Irradiance	taub		0.510	0.599	0.663	0.598	0.500	0.448	0.438	0.442	0.480	0.553	0.492	0.492
		taud	2.004	1.812	1.702	1.890	2.213	2.367	2.392	2.365	2.245	1.994	2.139	2.072
	Ebn at Noon		780	728	697	744	808	843	853	858	829	757	792	784
		Edh at Noon	171	216	247	204	146	123	121	125	141	177	148	156
All-Sky Solar Radiation	RadAvg		4.83	5.27	5.51	5.99	5.73	5.42	5.14	5.01	4.83	4.90	4.92	4.75
	RadStd		0.30	0.20	0.33	0.25	0.33	0.28	0.25	0.30	0.23	0.27	0.23	0.17

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	+0.23	+0.26	N/A	N/A	+100	+98

Nomenclature: See separate page